INFLUENCER EFFECTIVENESS MARKETING

Group Number

2 - 9

Team Member

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EXECUTIVE SUMMARY

This study evaluated influencer marketing effectiveness for Bath & Body Works' new perfume line by analyzing 250 Instagram posts from 5 mid-tier influencers.

Bottom line:

WHO posts and **HOW the image looks** drive engagement (not sentiment or topic). Recommend influencers to create content based on their natural lane.

MANAGERIAL IMPLICATIONS

01 Prioritize creator–brand fit

02 Visual richness is more important than safe aesthetics

03 Limit hashtags

WHAT WORKS?

Visual Richness (More objects in frame → more likes)

Rich, scene-based imagery activates sensory imagination → critical for fragrance marketing.

Creator Identity (Influencer Fixed Effects = strongest model signal)

Audience loyalty to the creator outweighs any single content tactic. → Pick the right person first as the influencer

Saturation: U-shaped (Very low OR very high works best)

Safe images underperform → Recommended to go bold or go minimal and avoid the forgettable middle.

WHAT DOESN'T WORK?

Hashtags (More hashtags = fewer likes & comments)

Over-tagging signals low-quality content and suppresses organic discussion → Keep them minimal.

Sentiment scores are not significant

Being positive is the baseline, not a differentiator. Tone alone does not predict engagement.

Topics fade with controls

Topic theme itself is not as important as niche focus. Topic effects disappear once creator differences are accounted for.

THINGS TO NOTE

Perfume = Complex Contagion

Multi-source reinforcement drives purchase → Mid-tier creators (10K–500K) generate the strong local ties needed to turn awareness into buying intent.

Mid-Tler Influencer Superior Engagement

Mid-tier influencers demonstrated superior engagement quality compared to macro-influencers, supporting the authentic, lifestyle-integrated fragrance content.

Limitation

50 posts/creator, 1 platform, no sales data.
Results play as guide for creative strategy but not as proof of business impact. Validate with A/B testing and conversions.

MID-TIER INFLUENCERS FIT FRAGRANCE STORYTELLING BEST

GOAL

Evaluate influencer marketing effectiveness for Bath & Body Works' (B&BW) new perfume line

STRATEGY

Leveraging Nano & Mid-Tier Influencers to trigger "Complex Contagion" — where social reinforcement from multiple relatable sources drives actual purchase behavior rather than just brand awareness.

Why Nano & Mid-Tier?

01 The Inverted-U engagement

Interaction efficiency peaks in the Mid-tier group. More than 1M followers, personal connection is diluted (Wies et al., 2022).

02 Maximizing ROIS

This group has "Strong Local Ties", which helps reduce access costs but increases conversion rates (Beichert et al., 2023)

03 Perfume = Complex Contagion

Fragrance needs social reinforcement before purchase — multiple sources must validate. Mid-tier influencers create strong local ties that drive actual behavior change (Centola et al., 2007)

Influencer	Followers	Engagement Rate	Type of influencers	Reason for choosing
@she.adoress	15.7K	9.49%	Fresh influencers	Fewest number of followers (nano influencers) → more variation in the data to analyze engagement. Higher engagement rate → strong local ties (personalized content) with followers
@professor.perfume	105K	1.16%	Fresh influencers	Mid-tier and fresh influencers → acquire new customer segment
@ijesimonn	139K	0.18%	Fresh influencers	Having fashion-forward style → effective for targeting customers who love aesthetic packaging Being a fragrance brand owner → have deep expertise for perfume and scent
@funmimonet	312K	0.81%	Familiar influencers	Mid-tier and familiar influencer → reinforce existing customers
@dogfragrance	346K	0.69%	Fresh influencers	The content focuses less on direct scent advertising but integrates into life (social/lifestyle) → broad audience fit for customers who care for authentic and honest review

WE BUILD VARIABLES TO EXPLAIN ENGAGEMENT DRIVERS

METHODOLOGY

1. DATA COLLECTION
2. IMAGE ANALYSIS
3. TEXT FEATURE ANALYSIS
4. TOPIC MODELING
5. DATA INTEGRATION & STORAGE
6. REGRESSION ANALYSIS & INTERPRETATION

VARIABLES GENERATED

Feature group	Variables generated	What it captures	Why included
Emotional tone	AFINN Score, NRC Emotion Variables , Vader compound	Positivity and emotional tone of the caption	To test whether positive tone, uplifting emotion, trust, and social-media-aware sentiment affect engagement
Caption style	Hashtags, mentions, questions, exclamations, emojis, caption	How the message is written and structured	To test whether stylistic choices affect engagement
Content themes	5 LDA topics	Main themes discussed in captions	To identify which types of content are more engaging
Image Features	Brightness, saturation, contrast, warmth	How the image looks	To test whether visual style affects performance
Visual richness	Number of objects	How detailed the image is	To test whether more number of visuals attract more engagement
Controls	Post age, influencer fixed effects	Timing and creator-specific baseline differences	To separate content effects from creator/timing effects
Outcomes	Likes & comments count	Audience engagement	Used to measure campaign performance

ENGAGEMENT VARIES WIDELY ACROSS INFLUENCERS

01 Quality over quantity

she.adoress writes the longest and receives the second most discussions (comments) despite fewer followers. → "deep engagement"

02 The Power of Lifestyle

funmimonet proves that when combining perfume with lifestyle (not only technical reviews) → the number of interactions will explode

03 Content Expansion

The appearance of skin and hair in the Top 5 keywords shows that the perfume market on Instagram today is not independent but associated with the Beauty & Self-care ecosystem.

Influencer	Dominant themes	Avg Likes	Avg Comments	Avg Words in Captions	Style
funmimonet	**fragrance, sephora, life, year**	**4187**	**229**	**40.4**	**Lifestyle + beauty shopping**
ijesimonn	outfit, shopping	548	38	38.2	Fashion-forward, call-to-action
dogfragrance	perfume, time, daily	534.41	21.98	40.4	Social/lifestyle, less perfume focus
professor.perfume	scent, smell, collection, honey	1486	29	47.5	Review perfume used daily, friendly lifestyle
she.adoress	**skin, texture, ingredient, scalp**	**434**	**63**	**59**	**Skincare-heavy, educational**

Figure 1: Influencers' caption wordcloud

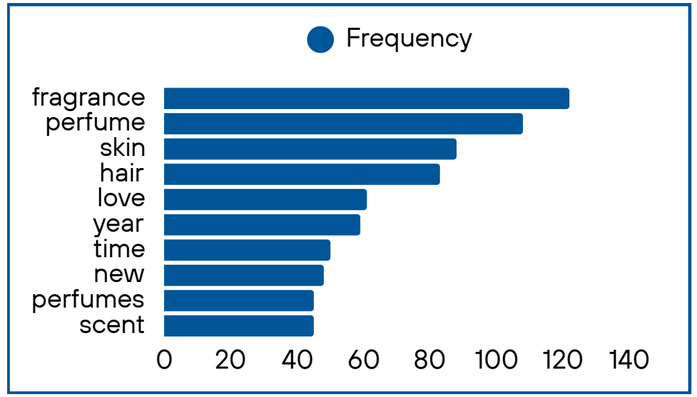


Figure 2: Top 10 words in caption

1425.30
Mean Likes
Median 614.50

1960.06
Std Dev (Likes)
High post-level variance

76.68
Mean Comments
Median 39.50 → right-skewed

46.4
Mean Caption Words
Median 36.5 → long-tail

SENTIMENT HAS LITTLE IMPACT ON ENGAGEMENT

01 TESTED 4 LEXICONS

Screened AFINN, NRC, VADER & Bing

02 KEPT THE MOST COMPLEMENTARY

Dropped Bing overlapped with AFINN conceptually and it adds less nuance. (Refer to the Boxplot in the appendix)

03 KEY FINDING

Across all 3 lexicons, the vast majority of captions scored positive, leaving little variance for engagement models to work with.

04 POSITIVITY ≠ ENGAGEMENT

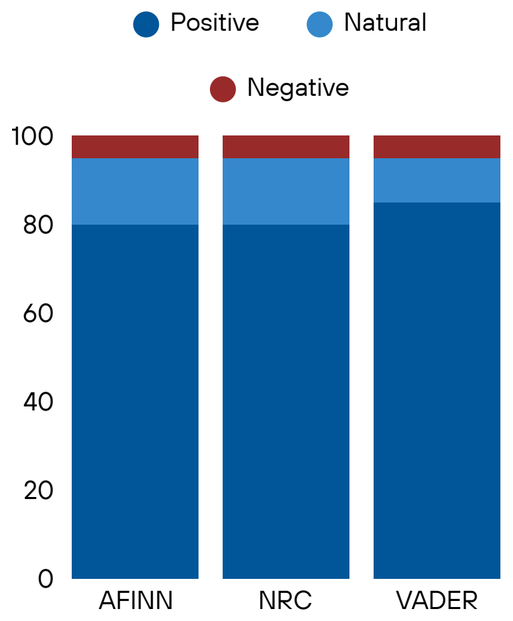
Despite captions skewing positive, sentiment had only a weak correlation with likes or comments.

LEXICON SELECTION

AFINN	✓ Retained	Strongest variation in sentiment intensity — long positive tail helps distinguish highly positive captions from average ones.
NRC	✓ Retained	Widest emotional spread — useful for separating uplifting emotions (joy, trust) from rarer negative tones.
VADER	✓ Retained	Built for social media — captures slang, emojis, and punctuation emphasis that pure text lexicons miss.
BING	✗ Dropped	Overlaps conceptually with AFINN, adds less nuance than NRC, and lacks social-media sensitivity of VADER.

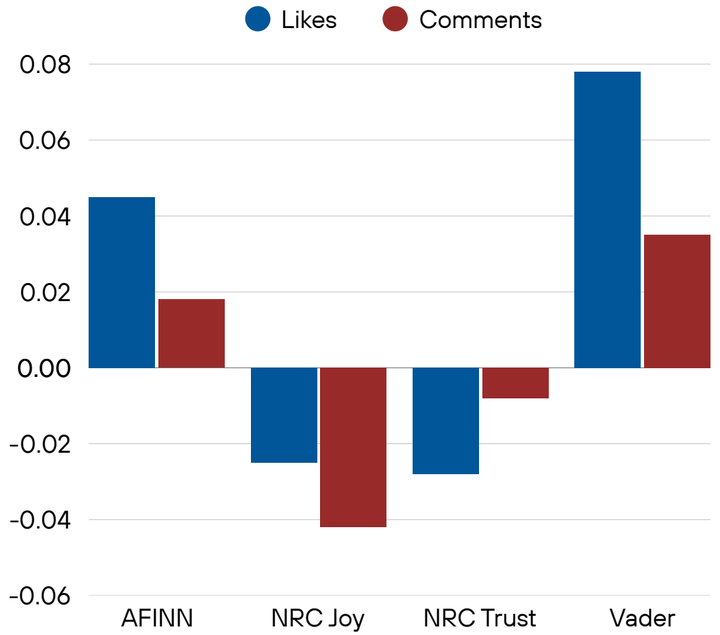
SENTIMENT DISTRIBUTION

Most captions are classified as Positive *across all lexicons*



CORRELATION WITH ENGAGEMENT

Sentiment scores show weak, inconsistent relationships with Likes & Comments



TOPIC STRUCTURE REVEALS DISTINCT INFLUENCER CONTENT LANES

01 5 CLEAR CONTENT THEMES

LDA across all posts identified five distinct thematic clusters in fragrance & beauty content. (Check Metrics Here and topics here in the appendix)

02 THEME ACROSS EACH CREATOR

Every influencer shows a dominant topic probability — signaling strong personal content specialization.

03 TOPIC FIT > ANY SINGLE THEME

In regressions, topic effects weaken once creator & post characteristics are controlled, showing that fit matters more than theme.

04 STRATEGY: BRIEF WITHIN THEIR NATURAL LANE

Forcing a single messaging style is not recommended. Each creator should promote B&BW through the content mode they already own.

TOPIC DEFINITION

T1 Emotional & Aspirational Lifestyle
Reflective, identity-driven fragrance content
Keywords: year, love, feel, dream

T2 Everyday Perfume Praise
Accessible fragrance appreciation and favorites
Keywords: perfume, scent, favorite, wear

T3 Skincare & Haircare Routines
Beauty-care content adjacent to fragrance
Keywords: skin, hair, serum, scalp

T4 Niche Fragrance Notes & Reviews
More technical note-based perfume evaluation
Keywords: fragrance, note, vanilla, niche

T5 Luxury Layering & Trending Aesthetic
Premium, stylized, trend-linked beauty/fragrance content
Keywords: body, aesthetic, luxury, layer

TOPIC PROBABILITY BY INFLUENCER

Dominant topic shown in bold and each creator occupies a distinct content lane

Influencer	T1 Emotional Lifestyle	T2 Everyday Praise	T3 Skincare & Haircare	T4 Fragrance Revies	T5 Luxury & Aesthetic	Dominant Topic (Share)
@dogfragrance	0.353	0.101	0.014	0.51	0.023	T4 (53%)
@funmimonet	0.744	0.073	0.04	0.075	0.068	T1 (82%)
@ijesimonn	0.742	0.089	0.03	0.035	0.105	T1 (88%)
@professor.perfume	0.274	0.599	0.011	0.039	0.077	T2 (66%)
@she.adoress	0.085	0.097	0.441	0.056	0.322	T3(50%)

LIKES DRIVEN BY VISUALS, NOT CAPTIONS OR TOPICS

01 TOPIC EFFECTS ARE NOT ROBUST

Topic only has meaning in a simple model.
When adding controls → effect disappears

02 STRONGEST FACTOR

The number of objects in the photo ↑
→ likes increased significantly
→ Visual content (with many objects) is more important than text.

03 NEGATIVE IMPACT

- A lot of hashtags → reduce likes
- Old post → fewer likes

→ **Over-optimization (spam hashtag) is ineffective**

04 NONLINEAR EFFECT (U-SHAPE) + INTERACTION

- Both low and high levels of saturation can outperform moderate levels
- With niche topics → object count is more important

MODEL EVIDENCE ACROSS SPECIFICATIONS

strong positive · strong negative · ~+ marginal positive · ns - not significant

Predictor	Raw Topics	Log Topics	Full Controls	Non- linear	Interaction	Final Verdict
Topic 1	+	+	ns	ns	ns	Not robust after controls
Topic 2	~+	+	ns	ns	ns	Initial effect disappears
Topic 3	ns	ns	ns	ns	ns	No evidence
Topic 4	ns	–	ns	ns	direct ns	Neg. simple log model
Sentiment (VADER)			ns	ns	ns	No contribution
Object Count			+	ns	+	Stronger driver than captions
Hashtag Count			–	–	–	Consistently neg.
Time Elapsed			–	ns	–	Older → fewer comments
Brightness			~+	weak curve	+	Weak nonlinear
Saturation			ns	U-shape	ns	Key nonlinear
Influencer FE			strong	strong	strong	Major source of variation
Topic 4 × Object count					+	Object richness matters more with Niche Fragrance Topic

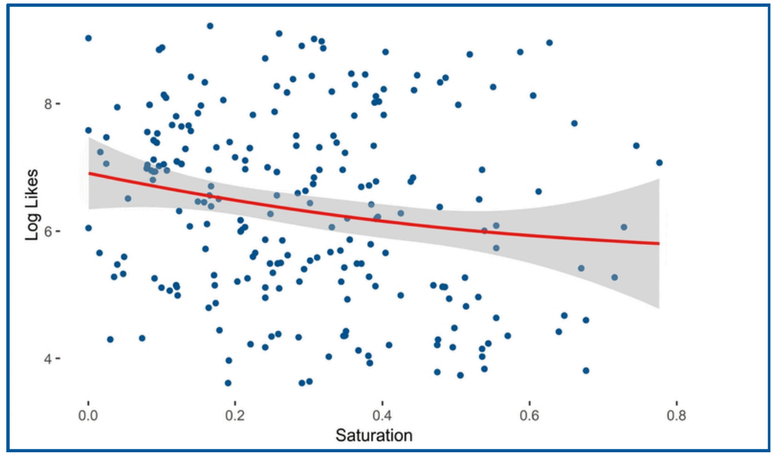


Figure 1: Quadratic Effect of Saturation on Log Likes

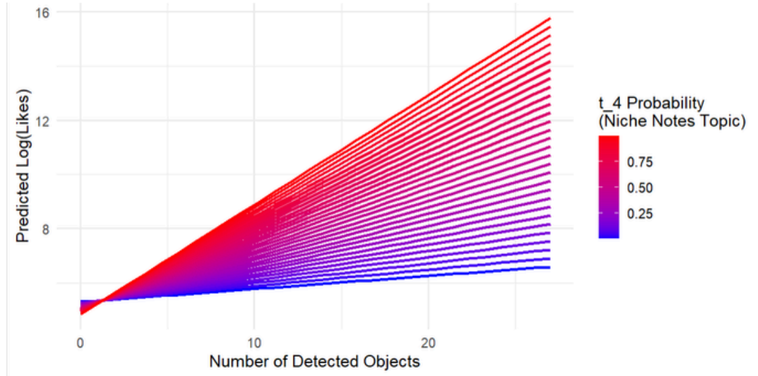


Figure 2: Interaction Niche Notes Topic x Number of Objects

COMMENTS WEAKLY DRIVEN, VISUALS IS STILL SLIGHTLY CRUCIAL

01 TOPICS & SENTIMENT

Topic effects appear in simple models but become non-significant once creator & post characteristics are added.

02 OBJECT COUNT

Object count has an impact but is weak. More objects → comments increase slightly but the effect is weaker than likes.

03 NEGATIVE IMPACT

- More hashtags → fewer comments
- Old post → reduce comments

➔ Same pattern as likes

04 SATURATION IS THE MOST IMPORTANT FACTOR (U-SHAPE)

Nonlinear relationship (U-shaped) with comments. Both low and high levels can outperform moderate levels

MODEL EVIDENCE ACROSS SPECIFICATIONS

strong positive · strong negative · ~+ marginal positive · ~+ marginal negative · ns - not significant

Predictor	Raw Topics	Log Topics	Full Controls	Non- linear	Interaction	Final Verdict
Topic 1	ns	ns	ns	ns	ns	No evidence
Topic 2	ns	~–	ns	ns	ns	Weak early effect
Topic 3	ns	ns	ns	ns	ns	No evidence
Topic 4	ns	–	ns	ns	ns	Neg. simple log model
Sentiment (VADER)	ns	ns	ns	ns	ns	No contribution
Object Count			ns	ns	~+	Weak positive
Hashtag Count			–	–	–	Consistently neg.
Time Elapsed			–	ns	–	Older → fewer comments
Saturation			ns	U-shape	ns	Key nonlinear
Influencer FE			strong	strong	strong	Major source of variation
Interactions					none	No interaction effect

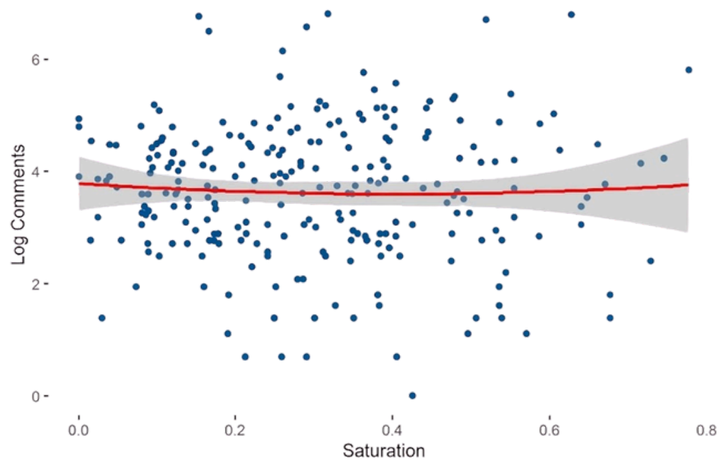


Figure 1: Quadratic Effect of Saturation on Log Comments

OVERALL RESULT DISCUSSION

2 x 2 Metrics

	STRONG EFFECTS	WEAK EFFECTS
✓ **EXPECTED**	i *Post age (time_c) negative* *Older posts get fewer likes and comments — recency signals freshness and relevance.* ii *Visual complexity (object_c) positive for likes and comments* *Visual complexity positive for likes and comments in the interaction comments model.* iii *Influencer fixed effects strong* *Engagement differs substantially by creator — who posts matters far more than what they post.*	i *Brightness positive for likes* *Direction is plausible, but the effect is only marginal or model-dependent — consistent with expectations but not robust.* ii *Topic 4 × object count for likes* *Richer visuals help certain content types (niche fragrance reviews). Expected interaction — but modest in size.*
✗ **UNEXPECTED**	i ***Hashtag count negative*** *for likes & comments* *Counterintuitive as one expects hashtags to boost discoverability. Instead, over-tagging suppresses engagement.* ii *Saturation has a U-shaped effect* *Both very low and very high saturation drive more likes and comments. This is a clear nonlinear finding and more surprising than a simple positive/negative effect.*	i *Topics lose significance in full models* *topics mattered in simple models, but not after controls and influencer FE.* ii *Sentiment variables remain non-significant* *AFINN, NRC, and VADER do not robustly predict engagement.* iii *Most interactions are not significant* ***t4_c × hashtag_c*** *and* ***brightness_c × object_c*** *are weak/not supported.*

FINAL RECOMMENDATION & NEXT STEPS

FINAL RECOMMENDATION

Prioritize mid-tier influencers (50K–300K followers)
who naturally integrate fragrance into lifestyle content rather than hard-selling.

Focus on specific themes
Niche Notes & Emotional Lifestyle themes (Topic 1 & 4) with detailed, reflective captions.

Focus on specific image elements
Encourage visual richness (moderate number of objects) and low or high brightness/saturation.

Hashtag usage
Limit hashtag usage to avoid negative effects on engagement.

Leverage complex contagion strategy
Multiple trusted voices validating the perfume experience rather than single influencer promotion.

Influencer	Key Strength	Content Focus	Recommendation
@she.adoress	High comments + long captions	Skincare + layering (Topic 3)	Use educational storytelling with Joy & Trust; emphasize scent–skin layering
@professor.perfume	Balanced engagement	Everyday perfume (Topic 2)	Position as friendly educator; focus on Trust & Aspiration in daily scent stories
@ijesimonn	Strong aesthetics + fashion	Lifestyle & outfit (Topic 1 & 5)	Combine fragrance + fashion visuals; use Joy & Aspiration for appeal
@funmimonet	Highest likes	Lifestyle integration (Topic 1)	Lead as lifestyle ambassador; focus on Joy + emotional daily use
@dogfragrance	Authentic audience	Social / real-life content	Shift to "perfume in real life" storytelling; build Trust & Aspiration

NEXT STEPS

Testing strategy
Test the recommended mid-tier influencers with A/B campaign formats.

Analysis expansion
Expand analysis to include comment sentiment and share rate.

Content guidelines development
Develop content guidelines for influencers focusing on scent storytelling and lifestyle integration.

Limitation

Cross-sectional data
The study captured posts at a single point in time, making it difficult to observe long-term engagement trends.

Small sample size
Only 50 posts per influencer were analyzed, which may limit the generalizability of the findings

Lack of conversion data
Engagement metrics (likes and comments) were analyzed, but actual purchase behavior or sales conversion could not be measured

Platform limitation
Analysis was conducted only on Instagram; results may not fully apply to other social media platforms such as TikTok or YouTube.

APPENDIX - (1/3)

Lexicons Differ in Intensity, Emotional Breadth, and Positivity Bias

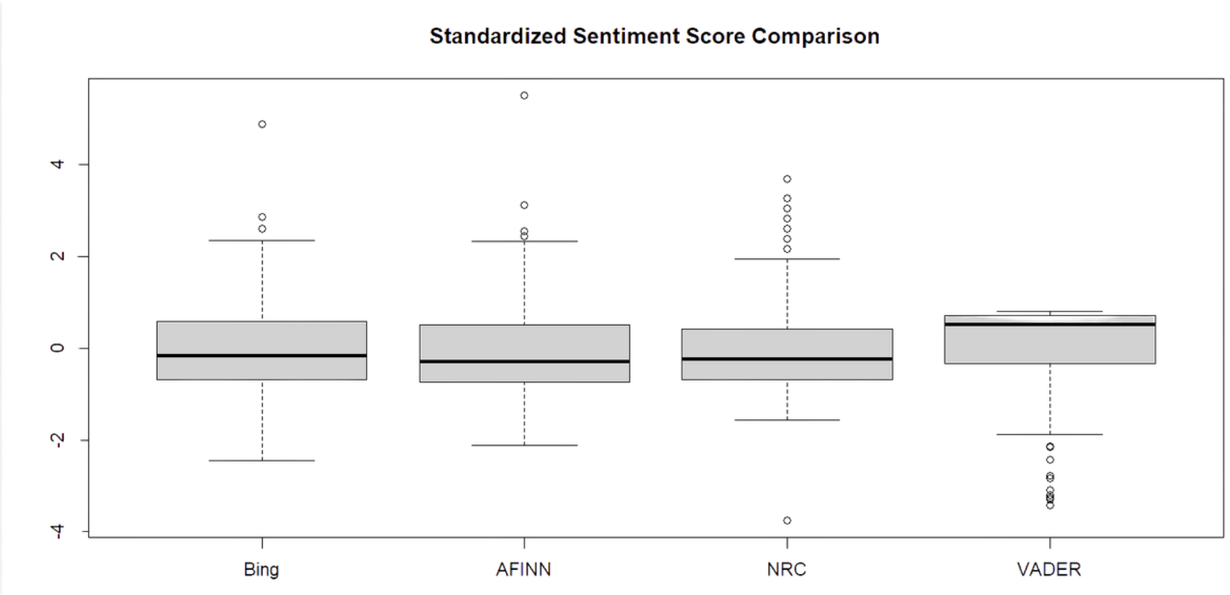


Figure 1: Boxplot comparison for Sentiment Lexicon

Main insights

- Bing is centered close to neutral and shows moderate variability, making it a useful baseline polarity measure but not the most distinctive lexicon.
- AFINN has the longest positive tail and several high outliers, indicating that it captures strong intensity differences better than the other methods.
- NRC shows the widest overall spread, reflecting greater variation in emotional content and making it useful for identifying specific emotional tones such as joy and trust.
- VADER has the highest median and the tightest central cluster, suggesting that most captions are consistently positive when social-media-aware sentiment is considered.
- VADER also shows several low outliers, meaning that while most posts are positive, a small number of captions are flagged as sharply less positive when emojis, punctuation, and informal style are taken into account.
- Overall, the boxplot suggests that the four methods are not identical:
  - AFINN is strongest for intensity,
  - NRC for emotional breadth,
  - VADER for platform-native sentiment,
  - while Bing serves mainly as a simpler benchmark.

APPENDIX - (2/3)

Metrics Do Not Converge on a Unique Optimal Topic Number

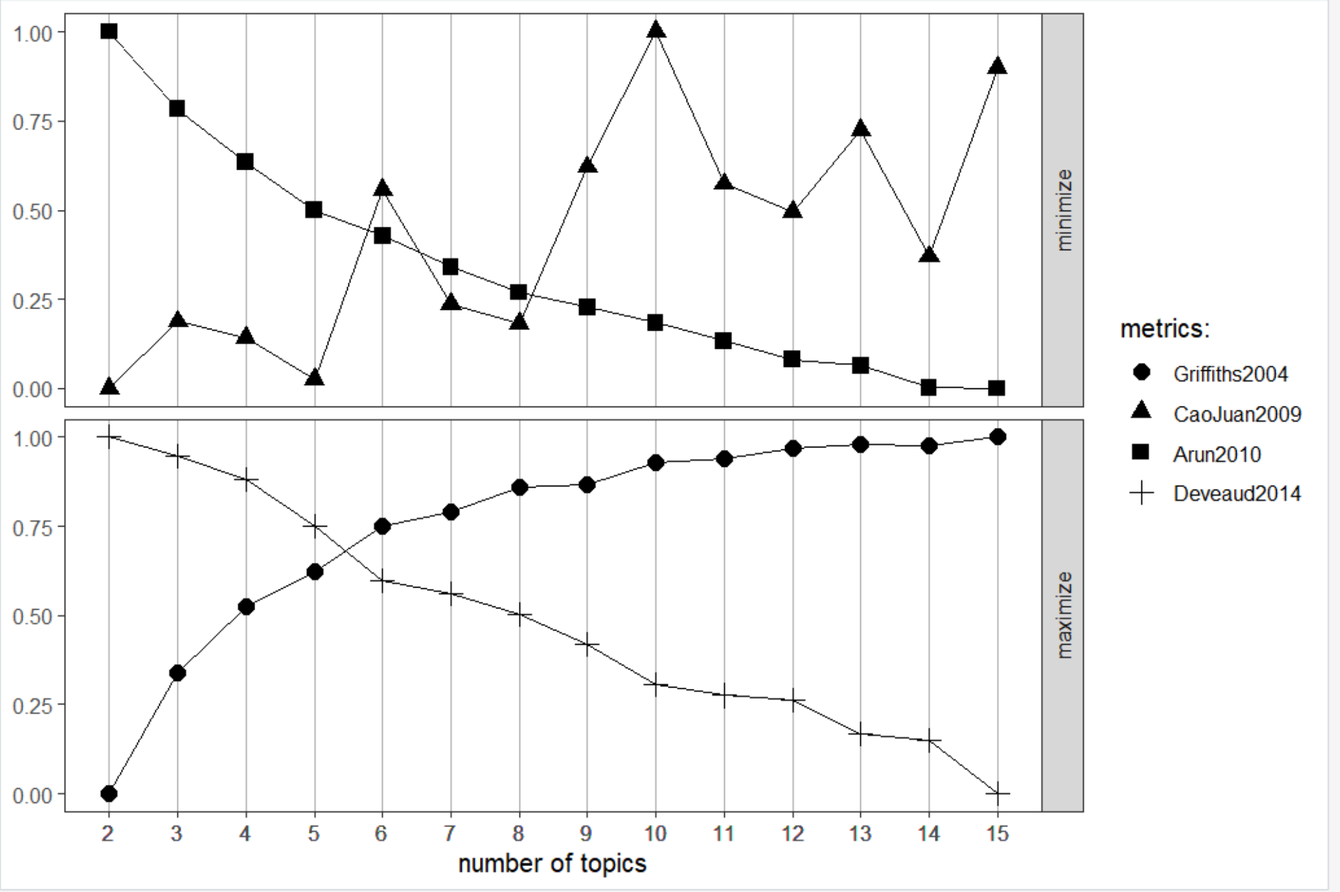


Figure 1: Topic Selection for LDA

LDA Topic Selection

Multiple topic coherence metrics were evaluated to determine the optimal number of topics.
While CaoJuan2009 indicates a clear minimum at k = 5 and Deveaud2014 favors lower values, the metrics do not converge on a single definitive solution. Griffiths2004, for instance, continues to increase with k and is known to overestimate the optimal number of topics.

Therefore, rather than relying solely on quantitative metrics, models within a reasonable range (k = 4–6) were examined qualitatively. The final selection was based on topic coherence and minimal overlap between topics.

Combining both quantitative evidence and qualitative interpretability, k = 5 provides the best balance between coherence and meaningful topic separation.

APPENDIX - (3/3)

Combining Metrics and Interpretability to Select k = 5

k	Coherence (mean)	R^2	Comment	Main topic pattern	Why it was not / was selected
4	0.184	0.1405	Too low	Broad themes: emotional/lifestyle, skincare, perfume notes, fashion/aesthetic	Too coarse. Key fragrance and emotional themes are merged, and the fashion topic feels less relevant to the perfume context. Coherence and R^2 are also relatively low.
5	**0.2057**	0.2068	**Highest coherence**	**Emotional lifestyle, everyday perfume praise, skincare/haircare, niche fragrance notes, luxury layering/aesthetic**	**Selected. Best balance of interpretability, coherence, and perfume relevance. Topics are clearly separated, meaningful, and not overly fragmented.**
6	0.177	0.1688	Drop in both	Similar core themes, but fashion/outfit and event/launch topics split out separately	Too narrow. Starts creating filler topics that are less perfume-specific and less managerially useful. Coherence drops.
7	0.1587	0.1764	Worst coherence	Multiple overlapping fragrance topics plus fragmented skincare/beauty subtopics	Too fragmented. Topics lose distinctness and become harder to interpret. Coherence is the weakest.
8	0.1752	0.2553	Good R^2, lower coherence	More niche separation, with very specific skin/hair and aesthetic subtopics	Too forced. Higher R^2 but lower coherence. Creates overly specific micro-themes with limited practical value.
9	0.1816	**0.2626**	Highest R^2, but coherence lower	Strong fragmentation into many overlapping fragrance and random micro-topics	Not suitable. Topics become too small, noisy, and repetitive, with reduced interpretability.
10	0.1877	0.2398			